

Project Demonstration Plan

FPGA-Accelerated Bitcoin Mining Using Double SHA-256

The purpose of this demonstration is to showcase the:

1. The HLS synthesis results of our optimised implementation
2. The FPGA-accelerated HW/SW Bitcoin miner visually running on the board
3. The results/performance of the optimised kernel running on the board
4. The results/performance of the ARM PS software baseline.

The demonstration will verify that both implementations produce correct results and will compare their execution performance.

1. Demonstration Setup

Hardware

- Kria KV260 FPGA development board
- Required connection between the host and KV260

Software

- Software-only Bitcoin miner
- FPGA-accelerated Bitcoin miner
- Vitis/XRT runtime environment
- Terminal for executing both implementations and displaying results

HLS synthesis results

- The resource utilisation of our optimised design
- The optimal initiation interval we achieved, $II = 1$, from our SHA_CHUNK

2. Software-Only Baseline Demonstration

Purpose

Demonstrate the operation and correctness of the baseline Bitcoin miner running entirely in software.

Information Shown

- Whether a valid nonce was found
- Resulting nonce
- The throughput, measured in MH/s
- TEST PASS/validation result
- Execution time

3. FPGA-Accelerated HW/SW Demonstration

Purpose

Demonstrate the operation and correctness of the accelerated Bitcoin miner using the FPGA implementation.

Information Shown

- Whether a valid nonce was found
 - Resulting nonce
 - The throughput, measured in MH/s
 - TEST PASS/validation result
 - Execution time
 - SPEEDUP comparison of the hardware and software.
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4. Correctness Validation

The results of the two implementations will be compared to demonstrate that hardware acceleration has not changed the functionality of the Bitcoin mining algorithm.

The following will be checked:

- Software and hardware implementations process the same input.
- Both implementations identify the expected valid nonce.
- The resulting hash satisfies the supplied target.
- Where applicable, the generated hash is compared against a known/reference result.